LESSON 8.3

Write and Solve Equations Using Multiplication or Division

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve one-step multiplication and division equations using inverse operations.

WHY THIS MATTERS Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Inverse operation *Spanish: Operación inversa*	Two math actions that undo each other, like × and ÷.	× and ÷
Multiply *Spanish: Multiplicar*	To add equal groups again and again.	3 × 7 = 21 means 3 groups of 7 equals 21
Divide *Spanish: Dividir*	To split into equal groups.	20 ÷ 4 = 5
Isolate *Spanish: Despejar*	To get the letter by itself on one side.	get p by itself
Coefficient *Spanish: Coeficiente*	The number in front of a letter, like the 3 in 3x.	the 3 in 3x
Solution *Spanish: Solución*	A number that makes the equation or inequality true.	x = 3 is the solution of 6x = 18 because 6 × 3 = 18.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

5 boxes each hold the same number of pencils, 45 in all. Solve 5b = 45.

1. Divide both sides by 5: 5b ÷ 5 = 45 ÷ 5.
2. Simplify: b = 9.
3. Check: 5 × 9 = 45.

Answer: Each box holds 9 pencils.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

Solve x ÷ 4 = 9. Is x multiplied or divided?

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The variable is multiplied by _____ , so I _____ both sides by it.
- My equation is _____ because each group holds _____ .
- The solution _____ means _____ in this problem.
- I checked my answer by multiplying _____ and _____ .

Word bank equation | coefficient | multiply | divide | inverse operation | isolate | both sides | equal groups | solution | check | undo | balance

WATCH OUT!

A common mistake

A common mistake in Solve Multiplication and Division Equations is applying the SAME operation again instead of the inverse

LESSON 8.3

Write and Solve Equations Using Multiplication or Division

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can solve one-step multiplication and division equations using inverse operations.

WHY THIS MATTERS Detective Park discovered that 3 identical locked boxes contain a total of 21 pieces of evidence. She writes the equation 3x = 21 to find how many pieces are in each box. How can she solve for x?

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

5 boxes each hold the same number of pencils, 45 in all. Solve 5b = 45.

1. Divide both sides by 5: 5b ÷ 5 = 45 ÷ 5.
2. Simplify: b = 9.
3. Check: 5 × 9 = 45.

Answer: Each box holds 9 pencils.

1 **Circle the letter of the best answer. Show how you know.**

Solve: 6x = 42

(A) x = 7
(B) x = 36
(C) x = 48
(D) x = 252

STEPS TO TRY
1. 6x means 6 TIMES x — the variable is being multiplied by 6.
2. Division undoes multiplication. Divide both sides by 6 to leave x alone.
3. Compute 42 ÷ 6, then check by multiplying your answer by 6 — do you get 42 back?

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Solve: n / 3 = 8

(A) n = 3

(B) n = 5

(C) n = 11

(D) n = 24

STEPS TO TRY

1. n is being DIVIDED by 3. Which operation undoes division?
2. Multiply both sides by 3 — the ÷3 and ×3 cancel each other, leaving just n.
3. Compute 8 × 3, then check: your answer divided by 3 should give 8.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Solve: 9m = 63

(A) m = 7

(B) m = 54

(C) m = 72

(D) m = 567

STEPS TO TRY

1. 9m means 9 times m — that's multiplication, even with no × symbol written.
2. Undo the multiplication with division: divide both sides by 9.
3. Compute 63 ÷ 9 (think: 9 times what equals 63?). Check by multiplying your answer back by 9.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

Solve: y / 7 = 5

(A) y = 2

(B) y = 7

(C) y = 12

(D) y = 35

STEPS TO TRY

1. y is split into 7 equal parts. To rebuild y, you need the opposite of dividing.
2. Multiply both sides by 7 so y stands alone on the left.
3. Compute 5 × 7, then confirm: your answer ÷ 7 should land back on 5.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide to solve an equation?

SENTENCE STARTER When I see ___ in the equation, I know to ___ both sides because the inverse of ___ is ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help